Experiment No.14

Title : To find out pH value of substances in everyday life with the help of universal indicator.

Apparatus : Test tubes, Test tube stand, Dropper, Beakers etc.

Chemicals and other materials : Universal indicator, Lime, Lemon juice, Fruit juice/Extract, Common salt, Detergent powder, Gypsum (Calcium sulphate), Soil, Milk, Banana, Food colour etc.

Note : In addition to the above materials, also take into account various other materials in the house, dyes, oils, deodorant etc.

Procedure :

1. Take sample of each material in a separate test tube.
2. Put some water in each test tube and prepare solution as per the requirement.
3. Put 2 drops of Universal indicator in every solution.
4. Observe the colour change taken place in the solution in each test tube and compare it with the colour shades on the bottle on the universal indicator
5. Record the pH of each material from this.

Observation table :

Name of the Material	Colour produced	pH
Lemon juice	Red	2
Sugar	Green	7.0
Common salt	colourless	1.5
Detergent Powder	Blue	8.5
Soil	colorless	
Milk	orange	16.5
Gypsum		
Banana		
Lime	Red	2.5

Name of the Material	Colour produced	pH
Fruit Juice	Red	
Deodorant	colourless	
Acidrain	orange	5-6
	orange	6-0
Thoatpaste	Blue	
limewater	Blue	
	orangle	6.0
Blood	Blaishgreen	7.4
Grastric	Red	1.0

Inference :

1. Acidic materials (pH less than 7) ...Tomata-vinegar
2. Basic materials (pH more than 7) ...Solution of Backing soda.
3. Neutral materials (pH 7) Pure water, sugar, solution
4. ...Strong Basic material, 1m NaOH.

Multiple Choice Questions

1. pH is measure of concentration of .....
   A. ions B. O^+ ions C. hydrogen ions D. all the above
2. In pure water H^+ = ----- mol/litre
   A. 1x10 B. $1x10^7$ C. $1x10^{-7}$ D. $1x10^{-14}$
3. The pH scale was introduced by the scientist named.....
   A. Bunsen B. Sorensun C. Feming D. Proust
4. The most accurate method to determine pH is .....
   A. pH Paper B. Universal indicator C. Electric pH meter D. All of these
5. Identify the odd one out with reference to pH
   A. Blood B. Toothpaste C. Detergent D. Urine

Questions

1. Is it possible to determine a material in edible or not by using pH? How?

A salt is neutral when it is play value is 7 such all cl salt is made from a strong acid base play value salt made from acid and weak basic is less than 7

2. While making various food stuff, jaggery is used along with tamarind. Why is this so?

Just to mask some of the tartneses or sourness of acid present in the tamarind and to give us Hs a slight of flavours

VSPPHH